



**UWI School of Business and
Applies Studies Limited (trading as ROYTEC)**
136-138 Henry Street, Port of Spain

SENIOR SEMINAR BOOKLET II

Bachelor of Science Degree in Information Systems Management

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3. PROJECT REPORT STRUCTURE

- 3.0 TITLE PAGE** – The title page should identify the Title of the Programme, your name and student identification number, the name of your Faculty Advisor, the officially approved project case/title, and the date submitted.
- 3.0.1 TABLE OF CONTENTS** – List each section according to this Project Structure and corresponding page/s. Include a list of Appendices in this section.
- 3.0.2 EXECUTIVE SUMMARY** - Write a few paragraphs describing at a high level, the key elements of the project that are detailed throughout the Project document.

3.1 SECTION 1 - IMPLEMENTATION PLAN

This section contains text and diagrams/charts. Define milestones and Activity Schedules of the project solution/s, integrating three (3) key elements: deliverables, due date or duration, and critical dependencies. In the text section, briefly discuss the project schedule (e.g., timeline for major phases) and define the project milestones and baselines. Here you should also identify any critical task or resource dependencies that may significantly impact your ability to adhere to the planned schedule (e.g. if the client must provide certain IT resources or access to end-users in order for you to complete a task). *(If a Gantt chart is used, identify and schedule all early project activities and resources related to analysis and design and, more broadly, later activities such as testing, and installation. Your Gantt chart should include, at minimum, the following information: Activity ID and Name; Brief Activity Description; Start and Completion Dates; Team Member(s) Assigned; Resources Required; Schedule analysis etc).* Be sure to schedule team meetings and client reviews, that is, plan not only the activities required to deliver your intervention but also those activities required to manage your process. You should present estimates for the duration of the project; the resource requirements required; the material and equipment resources in terms of quantity and duration.

You should refer readings on defining planning for your project, as documented in Chapters 4 and 6 of the recommended text.

3.2 SECTION 2 – PROTOTYPING & LOW LEVEL DESIGN

Prototyping is a method of obtaining early feedback on requirements by providing a model of the expected solution before fully building it. This technique is useful for your role as the business analyst to learn and discover what is valuable from the perspective of the stakeholder. Because prototypes are tangible stakeholders are able to visualize and experiment with a model of the solution rather than discussing abstract representations of the requirements.

A primary goal of building a prototype is to test designs (and solution ideas) before creating real solutions. Your solution's success is directly related to whether you test it or not. Without any doubt, your design will be tested when the solution becomes available on the market and people begin using it. If this is the first-ever testing, there's a high likelihood of negative feedback from stakeholders. Therefore, it's always better to collect feedback during the low-risk research phase, and before final release.

You are required to complete low level designs and pseudocode for the models completed in Senior Seminar I. For example in Senior Seminar I, you were expected to at least show a L1 feature model. Most feature models will have three or fewer levels of features such as Level 2 (L2) features, and so on which is a requirement in in Senior Seminar II.

You are also required to identify and document any changes to your analysed requirements and design section of the 3rd draft and final project report.

You should refer readings on prototyping for your project, as documented in Section 6.6 of the recommended text.

3.3 SECTION 4 – 3RD DRAFT AND FINAL REPORT

Iterative updates to the approved solution proposal in Senior Seminar I based on unknowns being identified and further insights into stakeholder requirements etc. as the solution is being implemented.

3.4 SECTION 3 - EVALUATION

Evaluation consists of business analysis activities performed to verify and validate the prototypes and the solution. Evaluation determines how well the solution meets the business needs expressed by stakeholders. Solution and prototype evaluation activities provide the ability to assess whether or not a solution has achieved the desired business result. Factors to consider:

- What goals and objectives does the evaluation confirm?
- Does the solution have built-in measurement capabilities for the evaluation criteria?
- Consider an evaluation acceptance criteria
 - Management approval
 - User evaluation form
 - Verification activities for functional and non-functional requirements
 - Validate solution results (testing).

You should refer readings on prototyping for your project, as documented in Section 6.12 and 6.13 of the recommended text.

3.5 SECTION 5 - CONCLUSION

State the key learning/s that occurred from your work/project experience. These factors include relevant government or industry standards, organizational culture, politics, code of conduct, ethics and perceptions.

Project Checklist

Implementation

- Statement of whether or not the implementation achieves the design and if not, why not.
- Explanations of difficulties encountered, and how you resolved them.
- Explanation of tools used to support implementation and how they were used.
- Explanation of how technology was used to implement aspects of the design.
- Descriptions of files, data base tables, data structures, screen shots, etc.
- Identify any major changes from your Senior Seminar I project via **change requests** etc.

Evaluation Plan

- What does this evaluation strive to achieve?
- What is the purpose of this evaluation?
- How will findings from the evaluation be used?
- Who are the stakeholders for this evaluation?
- What role did they play in developing this individual evaluation plan?
- How do you plan to engage these stakeholders when implementing the individual evaluation plan (e.g., participate in collecting data, help to interpret findings)?
- Will there be internal, external, or both type of evaluation activities?
- What are the solution's intended outcomes (intended outcomes are short-term, intermediate, or long-term)?
- What do you ultimately want to change as a result of your activities (long-term outcomes)?
- What is the design for this evaluation? (e.g., focus group, questionnaires, sample form)
- Why was this design selected?

Testing

- Quality assurance checking of system models.
- Test strategy and test plan.

- Creation of appropriate test data.
- Logs of test runs.
- Comments on the results of testing.
- User acceptance testing.
- Results from unit, integration and system testing.
- Expected vs. actual results for functional requirements.
- Expected vs. actual results for non-functional requirements.
- A/B Testing (also known as split testing) is a form of quantitative analysis comparing two different versions of a product (e.g. two different types of landing pages)
- White-box and black-box testing

Solution Demonstration

- To see the software application working and:
 - Assess whether it meets all the stated system requirements.
 - Assess the quality of the user interface, and navigability.
 - Assess the degree of validation checks implemented.
- To verify the authorship of the product.

Also

- Overall screen design –well thought out? Consistent style used?
- Screen layout –good use of menus, buttons, positioning of fields, and so on?
- Screen navigation –easy navigation between screens?
- Software functions –do they meet system requirements? How well do they work?
- Data validation –appropriate validation of user-entered data? Type and range etc?
- Forms –is there opportunity for auto-filling of some fields?
- Overall report design - well thought out? Consistent style used?
- Report content –contains everything it should?
- Report layout –bespoke design? Generated from package? Customised?
- Algorithms, triggers, preconditions, post conditions, normal flow, alternative flows, exception flows and integrations.

Readings on demonstration tips and the skills needed to present and defend your project effectively in oral form for your project refer to Chapter 9 of the recommended text.

Quality of Report

- All the required sections: abstract, introduction, discussion of design issues, technical documentation and so on.
- Sections have clear introductions, well-structured paragraphs and a concluding paragraph that makes a connection to the next section.
- Proof read and spelling check before it is printed.
- Formatted in a professional manner with appropriate margins, font size and style, headings and so on.
- Does not contain basic errors in English grammar such as:
 - Commonly sounding words such as their and there.
 - Not writing complete sentences.
 - Spelling mistakes.

Initial Investigation

- Reasonable number of references given.
- Coherent section on findings of investigation.
- Correct referencing using MLA referencing notation.
- Implications of findings for project to be carried out.

System Evaluation and Conclusions

- A summary of what aims and objectives have been achieved.
- An explanation of how they developed their project.
- Conclusions drawn from their work.
- Discussion in which the project might have been tackled differently.
- Comment on what they were not able to achieve.
- Evaluation of their project planning and management.
- Comment on lessons learnt from undertaking the project.
- Recommendations for further development of the solution. Indicate how the project might be taken forward and any future research.
- Obtain signoff of the solution.
- Determine metrics.
- Assess limitations of the solutions.
- Recommendations to improve the solution performance.

Technical Documentation

- Requirements specification.
- Analysis models.
- Design models.

- Implementation model.
- Data structure design.
- Algorithm design.
- Flowcharts, etc.
- User interface design.
- Architectural design.
- Program design.
- Test data and test logs.
- Screen and report layouts.
- Code listings.
- User documentation e.g. installation guide.
- Error recovery.
- Visual programming.
- Storyboarding, mock-ups
- Wireframes.

Tools

- Mind mapping
 - <https://www.matchware.com/mind-mapping-software>
- CASE tools
 - Microsoft Visio
 - <https://www.dbdesigner.net>
 - <https://erdplus.com>
 - <https://www.draw.io>
 - <https://www.lucidchart.com/pages/>
- Prototyping tools
 - Screen generators.
 - Report writers
 - Application or code generators
 - Fourth-generation languages
 - Fourth-generation environment
 - <https://www.axure.com>
 - <https://moqups.com>
 - <https://mockingbot.com>

- <https://www.justinmind.com/free-wireframing-tool>
- <https://www.justinmind.com/ui-kits>
- <https://www.ux-app.com>
- <https://www.figma.com>
- <https://www.fluidui.com>
- <https://teleporthq.io>
- <https://www.softr.io/>
- <https://www.moxly.io/>
- Solution Hosting
 - www.000webhost.com
 - <https://infinityfree.net>
- App development
 - <https://developer.android.com/studio>
 - <https://visualstudio.microsoft.com/xamarin/>
 - <https://www.visualstudio.com/en-us/products/visual-studio-community-vs.aspx>
 - <http://www.jotform.com>
 - <http://www.phpform.org>
 - <http://www.coffeecup.com/web-form-builder-lite/>
 - <https://www.drupal.org>
 - <https://getbootstrap.com>
 - <https://apex.oracle.com/>
 - <https://databox.com/>
 - <https://teleporthq.io/>
 - <https://www.airtable.com/>
 - <https://bubble.io/>

Project Management

- Demonstration and evidence of project planning and control throughout the project:
 - Properly constructed Gantt chart with milestones and targets shown.
 - Ability to monitor actual progress against planned progress and to revise targets when necessary.
- Conduct in respect of:
 - The student arranging and attending meetings.

- The student participating interactively in meetings: discussing concepts, ideas and design decisions; reporting clearly and thoroughly on progress with evidence of work done.

Consideration of enterprise environmental factors and issues

- Consider the impact of organizational culture, politics, code of conduct, risk appetite and perceptions.
- States which aspects of the project raise ethical and/or legal issues and describe the issues.
- Discuss how these were dealt with and record the decisions made.
- Consider physical environment limitations such as working conditions or government regulations and standards influencing to the environment.

4. PROJECT PLANNING

1. The Project must consist of the student's account of his/her research findings as a result of carrying out the problem diagnosis, research and analysis, design interventions and evaluation. The intervention must be agreed with your Advisor, who will also work with you to determine scope, effectiveness and acceptance.
2. The Project must be of a satisfactory literary standard, free from typographical errors, void of plagiarism (see item 6 below) and must comply with the required academic protocol.
3. The Project must be typed using Times New Roman, 12 pitch point, page numbered, presented in **double line** spacing, on white bond paper and with suitable layout i.e. margins, allowing sufficient space for footnotes etc.
4. A table or a figure should appear in the text closely following the point where it is first discussed and should be listed by number e.g. Table 1.
5. The Project should be approximately **10,000-13,000 words** (excluding the list of references and appendices). The title page and index page containing the table of contents should not be numbered. Appendices should follow in sequence (numbered 1, 2, 3, etc.).
6. **Citations** - A key part of writing your Project is the documentation of references throughout the project and in the Bibliography. This means that you must give proper credit to the experts who have worked to provide you with the valuable information on which you have based your ideas and writing. **Failing to identify your sources can constitute plagiarism.** Plagiarism is the act of representing the work of another as one's own (without giving appropriate credit) regardless of how that work was obtained and submitting it to fulfil academic requirements. **Failure to reference your work is an academic offence and would result in your project being ungraded or receiving zero marks for that section of the project.** Plagiarism includes but is not limited to: *the act of incorporating the ideas, words, sentences, paragraphs, or parts thereof, or the specific substance of another's work, without giving appropriate credit, and representing the product as one's own work.*

You are responsible for making yourself familiar with the **Academic Regulations Book – Sub-section: Academic Offences – Plagiarism** which is available online at Document Downloads section under the Current Student tab at www.roytec.edu. Academic dishonesty also includes a failure on your part to keep your assignment papers out of the hands of other students who may misrepresent their origin. It is also suggested and recommended that you keep a record and/or rough drafts of written or other work until you have received your final grade.

The OWL website is a recommended source for information on the primary formatting guides for research papers and citations, it is found at: <http://owl.english.purdue.edu/handouts/research/index.html>.

For readings on defining planning for your project, refer to Chapter 4 of the recommended text.

STUDENT RESPONSIBILITIES

1. To consult with your Project Advisor throughout the Project term. You are encouraged to share your concerns or problems regarding the Project with UWI-ROYTEC. It is your responsibility to initiate communication with your Project Advisor (method to be determined with your Advisor i.e. individual or class meetings, email, telephone) and to seek clarification from UWI-ROYTEC where necessary. Your Project Coordinator at UWI-ROYTEC is Mr Name, Email: Telephone:
2. To identify a project/organisation of choice/an industry/a product on which you will be required to propose your project and to function professionally within all ethical and legal standards of behaviour.
3. To maintain confidentiality concerning all information gained from client organisations considering that it may be proprietary information. The student is not to discuss cases with anyone not professionally involved or sanctioned to need for the information. Any case record or material that could be discussed or used outside the student/advisor relationship must be amended or removed to protect the confidentiality of those involved. UWI-ROYTEC/the Project Advisor will not be held responsible for records that have been removed from any organisation for use in your Project. Project students must sign a Confidentiality Statement if required by any participating organisation.
4. Students must demonstrate the ability to integrate and apply knowledge, theory, skills, interventions, and professional ethics learned in the classroom to the project experience.
5. A First Draft Project document **must** be presented to the Project Advisor in accordance with the Completion Schedule.
6. To attend individual/group meetings with the Project Advisor and seek to develop a positive relationship with the Advisor.

7. All students are urged to participate actively in the process and seek to maximise the expert advice and supervision provided by your Project Advisor. Students are encouraged to ask the Project Advisor what is expected at every stage of the project and be prepared to receive critical and supportive feedback. Prior to meeting with your Advisor, students should formulate questions/concerns about the client organisation/project/product.
8. Students are responsible for notifying the Registrar registrar@roytec.edu of any change in status e.g. deferral of the Project to next year; leave of absence; withdrawal from the programme, inability to complete the Project on account of illness.

UWI-ROYTEC RESPONSIBILITIES

1. Provide each student with a Project Advisor, who will offer expert guidance for a meaningful learning experience as you work individually as opposed to the usual group/team structure.
2. Provide appropriate referrals and issue letters of authorisation to facilitating organisations or agencies which are relevant to your project.
3. Co-ordinate the progress of each student to ensure that he/she is progressing toward achieving the deadlines as set out.
4. Provide other appropriate support services that are compatible with the objectives of the Project.

PROJECT ADVISOR RESPONSIBILITIES

1. Your Project Advisor is also the appointed Examiner for your Project. He/she will provide guidance and direction to students for developing and evaluating an educationally directed Project.
2. Advisors will allow sufficient supervisory time to direct the learning experience of each student. This includes conducting regular meetings with students (individually or in organised groups) to clarify areas of difficulty, review work in progress, directing students in research methodology, sources of academic literature, and provide regular feedback to students.
3. Provide UWI-ROYTEC with a written evaluation of the student on forms provided by UWI-ROYTEC.
4. Review the draft Project document and provide feedback to students at least two (2) weeks prior to the final submission date.
5. Advisors are required to respond to inquiries from students and maintain communication with all students.

6. No Advisor is authorized to grant an extension of time, which may go beyond the final submission date as outlined by UWI-ROYTEC on the Completion Schedule as shown on page 1. This condition is necessary to ensure timely processing of nominations for the award of the Bachelor of Science Degree in Information Management Systems.

7. Advisors are **not** obligated to provide feedback to students who did not submit the first draft on the due date. Advisors will **not** mark the first draft in lieu of a final submission.

EXAMINATION REGULATIONS/ COURSE CREDIT

Students who wish to defer the Project must first proffer a reason in writing to the Registrar, UWI-ROYTEC at registrar@roytec.edu before **the 3rd week of term**. Please note that registration and administration fees will apply for your next sitting. Students who do not obtain a deferral/do not submit a Project will be marked **F** and will be required to register for the course at the next sitting. Please note that course fees (including registration and administration) will apply.

WRITING THE PROJECT

IMPLEMENTATION/ACTION PLAN

Design and present an implementation plan; it must establish timelines for the recommended intervention by determining course of action that can be implemented in the short, medium and long term.

Create a plan/outline that identifies:

- the key activities to be undertaken
- by whom
- time frames for implementation
- possible constraints

Create issue logs

Create communication plan

Create acceptance criteria

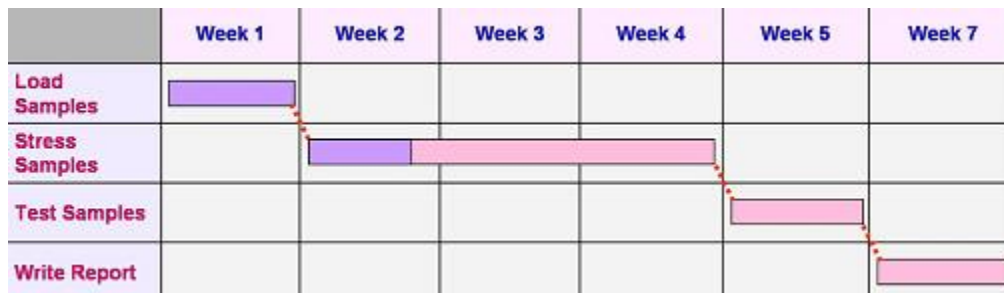
Create quality assurance plan

Create risk management plan

Your Implementation Plan (of the recommendations) must be shown, using a Gantt chart (a horizontal bar chart used in project management as a tool for graphically representing the schedule of a set of activities or tasks) along with possible resistance or support.

This section contains charts, tables and texts.

A simple Gantt chart



PROTOTYPING

It can be frustrating trying to elicit requirements from stakeholders who not know what the solution is that they are asking for. To address this challenge, prototyping helps stakeholders visualize the proposed solution to further define and meet their requirements. You will be required to produce and document two types of prototypes

- Low-fidelity prototype (1st solution prototype)
- High-fidelity prototype (2nd solution prototype)

Low-fidelity prototype (1st solution prototype)

Wireframes, mock-ups of user interface screens and reports to provide a visual representation of what the solution will look like and how it will function. Here are the basic characteristics of low-fidelity prototyping:

- Visual design: Only some of the visual attributes of the final product are presented (such as shapes of elements, basic visual hierarchy, etc.).
- Content: Only key elements of the content are included.
- Interactivity: The prototype can be simulated by a real human.

High-fidelity prototype (2nd solution prototype)

A representation of the final finished solution that is usable by stakeholders. Typically, this prototype has limited data, is restricted to a single interface computer device and has partial functionality. High-fidelity prototyping is performed in an iterative and evolutionary fashion. Reviewers can manipulate the screen, enter some data, and move from screen to screen to experience first-hand how the solution will work. With each successive prototyping session, more functionality is added or the existing functional is modified to achieve a higher level of quality. High-fidelity prototyping is performed using a programming language or a pseudo language of the solution to be demonstrated. The basic characteristics of high-fidelity prototyping include:

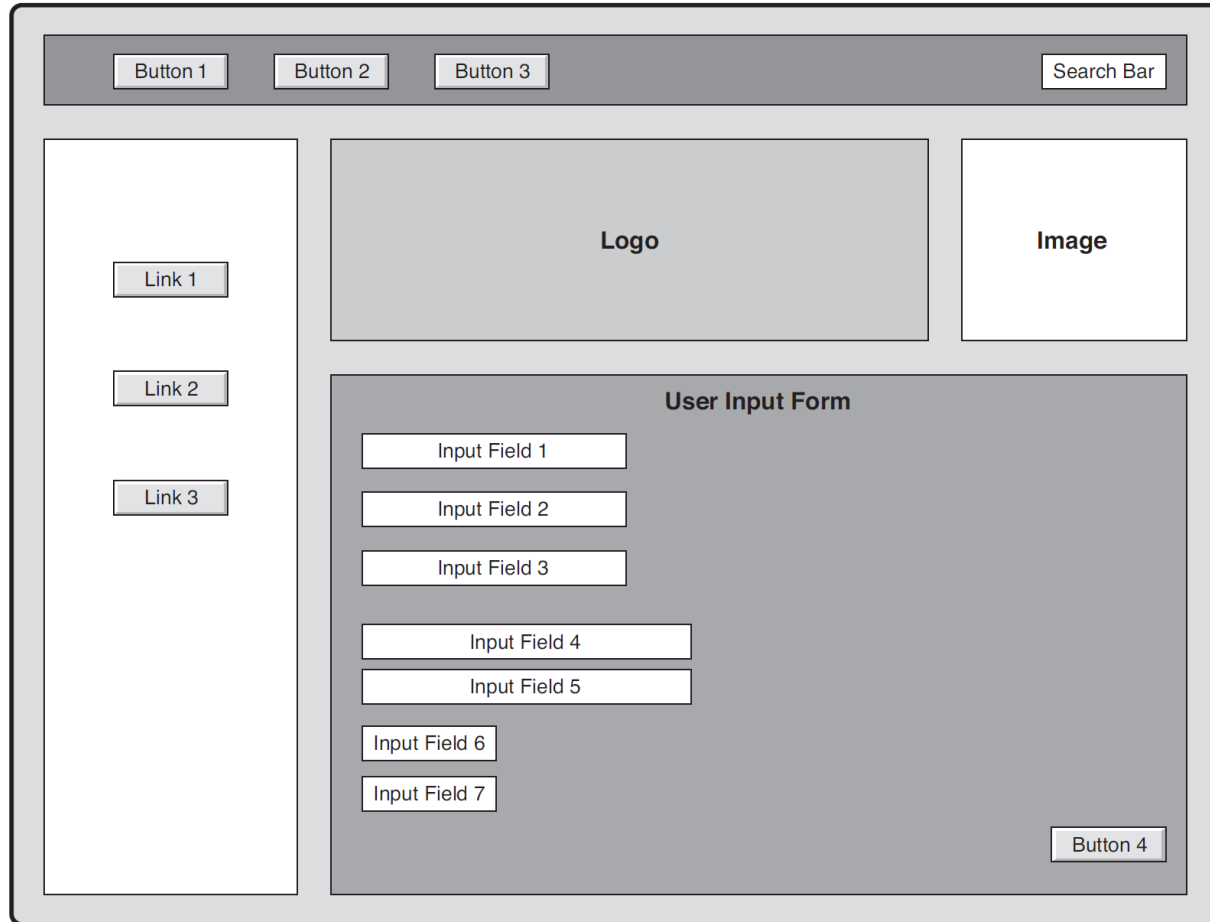
- Visual design: Realistic and detailed design — all interface elements, spacing, and graphics look just like a real app or website.
- Content: Designers use real or similar-to-real content. The prototype includes most or all of the content that will appear in the final design.
- Interactivity: Prototypes are highly realistic in their interactions.

Two examples of prototyping techniques are:

- **Storyboarding** is a prototyping technique showing sequence or navigation thorough a series of images or illustrations. In software development, storyboards use mock-ups to show navigation paths through webpages, screens, or other user interfaces. Storyboards are graphical representations that represent the sequence of events.
- **Wireframe** is a diagram representing a static blueprint or schematic of a user interface and is used to identify basic functionality. A wireframe separates the look and feel of a site from its function. It presents a stripped down, simplified version of the page, devoid of distractions. The purpose of the wireframe is to illustrate the flow of specific logical and business functions by identifying all entry and exit points and actions the stakeholders will experience. Wireframes drive **communication** and help to support evolutionary discovery of requirements. A typical wireframe contains:
 - Key elements such as header, footer, navigation, content objects, branding elements, and their respective locations
 - Groupings of elements such as side bars, navigation bars, and content areas

- Labelling, page title, navigation links
- Placeholders, content text and images.

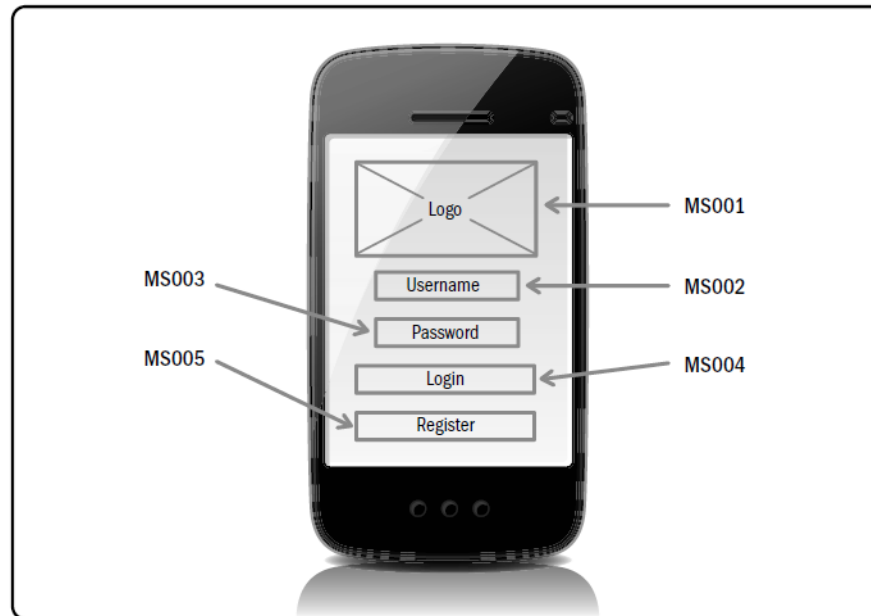
Example #1 of Wireframe



The wireframe illustrates a web page layout with the following components:

- Header:** A horizontal bar containing three buttons labeled "Button 1", "Button 2", and "Button 3" on the left, and a "Search Bar" on the right.
- Main Content Area:**
 - Left Sidebar:** A vertical column containing three links labeled "Link 1", "Link 2", and "Link 3".
 - Center:** A large rectangular area labeled "Logo".
 - Right:** A rectangular area labeled "Image".
- User Input Form:** A large rectangular area on the right side of the main content, titled "User Input Form". It contains seven input fields labeled "Input Field 1" through "Input Field 7", arranged vertically. A "Button 4" is located at the bottom right corner of this form area.

Example #2 of Wireframe



Each user interface element of the wireframe must be explained. For example, the table below models the display and behaviour requirements for the password button or MS003. It should be noted that wireframes must be traced directly to your high level and low level design models.

Example of Display-Action Response Model

UI Element: Password Fields	
ID	MS003
Description	A field for the user to enter password
UI Element Displays	
<i>Precondition</i>	<i>Display</i>
On login screen	Display field
No text entered	Greyed out password

Test entered	Privacy dots are displayed per character	
UI Element Behaviours		
Precondition	Action	Response
On login screen	Select field	Keyboard appears
Text entered	Enter additional text	Privacy dots per character appear

Report Table

Another technique is a report table which is an interface model that describes detailed requirements for a single report. Report tables contain both information about the report as a whole, such as the report name or decisions made from the report, and field-level information, such as which data fields are displayed and any calculations. Common attributes of the table include: name, description, decisions from the report, objectives, audience, trigger, data fields, data volume, frequency, display format and calculation. A report table typically accompanies a **prototype** of a report to show stakeholders what the report should look like. The first figure below shows an example prototype of a report and the other figure shows the report table model.

Example of Report Prototype

Mobile App Usage Report (MAUR)							
Reporting Period September 07, 2019-September 13, 2019							
Number of App Downloads	Number of Recipe Uploads	Store ID	Region	Number of Loyalty Program Transactions	Number of Products per Transaction by Users with Recipe Uploads	Average Number of Products per Transaction for Loyalty with Recipe Transactions	Average Number of Products per Transaction for Loyalty without Recipe Transactions
10,732	6907	0	Curepe	78563	4576	34	10
		23548	Arima	10,723	2,469	32	17
		54721	Port-of-Spain	6,093	1,098	33	15
		56098	San Fernando	15,497	543	30	12

Example of Top Level Elements in a Report Table

Element	Description
Unique ID (Unique identifier)	REP_MBL001
Name	Mobile App Usage Report (MAUR)
Description	The MAUR contains the total numbers of new app download, recipe uploads, and frequency of visits by app users
Decisions Made from Report (The business decisions that are made using the information in the report)	The MAUR will be used to determine if the Recipe Box feature is being utilized by mobile app users and if users are actually purchasing the products in stores
Objective (The business objectives support by this report)	Business Objective (improve profit per visit)
Priority (The priority to implement this report)	1 of 16
Functional Area (The business processes or areas that use the report)	Sales and Marketing
Related Reports (List of any other reports that have similar data)	Inventory Report (REP_INV001) and Sales Report (REP_REV001)
Report Owner (Business user(s) who own approval of the report requirements)	Marketing Manager
Report Users (Business user(s) who run or use the report to make decisions)	VP of Sales, VP of Marketing, Sales Representatives, Marketing Department, Mobile Service Representatives
Trigger (What triggers the report to run)	MAUR is generated automatically on Monday morning at 2am

Element	Description
Frequency (How often the report is generated and accessed)	Weekly
Latency (How quickly the report is delivered to users in relation to being requested)	MAUR is delivered within 2 minutes of being triggered and contains data for the previous week
Transaction Volume (How many transactions are pushed in the repository feeding the report)	Approximately 100,000 transactions and logged weekly
Security (Security on the report or fields that varies from security specified on the fields in the data dictionary)	Viewable by all sales and marketing members
Persistence (Expected saving of report settings between report sessions)	All settings are saved between report executions
Visual Format (How the report is visually displayed)	The Reporting period will be noted above the MAUR Matrix in the form “Month DD, YYYY-Month DD, YYYY” MAUR is a matrix with the following columns: Number of App Downloads, Number of Recipe Uploads, Store ID, Region, Number of Loyalty Program Transactions, Number of Loyalty Program Transactions by Users with Recipe Uploads, Average number of Products per Transaction for Loyalty with Recipe Transactions, Average Number of Products per Transaction for Loyalty without Recipe Transactions. The first row contains metrics for all stores The Store ID and Region will be “O” for the first row
Delivery Format	The report is sent automatically and periodically by system.

Element	Description
(How the report is delivered to users for viewing, any related functionality)	
Interactivity (Functionality within a report that allows the user to change views or other aspects of the data being displayed)	The MAUR can interacted in Excel format
Drilldowns (Links to other related reports or layers of this report with expanded data)	N/A

Low Level Design

In your low level design models, dialog, and written requirements, you want to use powerful words that describe the actions users take when using the system or solution.

Use powerful words that are:

- Concise
- Testable
- Easy for customers, developers, and business people and leaders to understand

Do NOT use words that are:

- Ambiguous
- Can mean many things

Sample List of Words to Avoid These words create a false sense of consensus on requirements and are not testable	Sample List of Words to Use
• Maintain • Manage • Administer • Handle • Process • Support • Facilitate • Track • Review • Monitor • Coordinate • Analyse	• Update • Edit • View • Display • Compare • Calculate • Determine • Print • Sort • Filter • Add • Confirm • Record • Enter • Attach • Authenticate • Adjust • Allocate • List • Notify • Request • Access • Import • Export • Merge • Refresh • Record • Send • Receive

Example

Vague requirement

- As an online shopping customer, I want to manage my profile.

Better discussed, modelled, and written requirements

- I want to create a profile.
- I want to view my profile.
- I want to update my profile.
- I want to attach a photo to my profile.
- I want to delete my profile.

Pseudocode

Pseudocode is a low level design technique for representing program logic. Pseudocode is similar to structured English, not programming language-specific, so you can use it to describe a software module in plain English without requiring strict syntax rules. Using pseudocode, a business analyst can describe program actions that can be implemented in any programming language. The below illustrates an example of pseudocode that documents a sales promotion policy.

Sample of a sales promotion policy:

- Preferred customers who order more than \$1,000 are entitled to a 5% discount, and an additional 5% discount if they used our charge card.
- Preferred customers who do not order more than \$1,000 receive a \$25 bonus coupon.
- All other customers receive a \$5 bonus coupon.

Pseudocode version of the sales promoting policy:

```
IF customer is a preferred customer, and
    IF customer orders more than $1,000 then
        Apply a 5% discount, and
        IF customer uses our charge card, then
            Apply an additional 5% discount
    ELSE
        Award a $25 bonus coupon
ELSE
    Award a $5 bonus coupon
```


Example of pseudo code for earlier flowchart algorithm

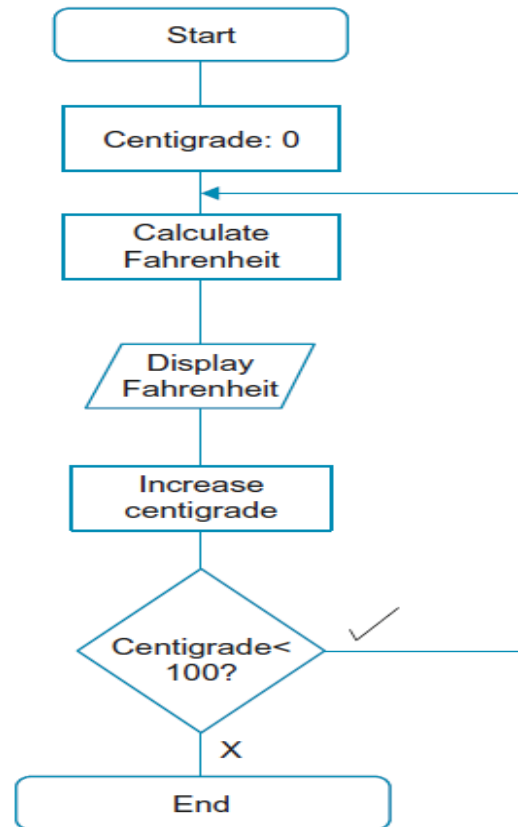


Figure 6.6 Example of a simple flowchart

```

cent = 0
do
    fahr = 1.8 x cent + 32
    print fahr
    cent = cent + 10
while (cent < 100)
  
```

USABILITY REPORT

The goals of usability testing include establishing a baseline of user performance, establishing and validating user requirements, and identifying potential design concerns to be addressed in order to improve end-user satisfaction and further align to user requirements.

The usability test objectives are:

- To determine design inconsistencies and usability problem areas within the user interface and content areas. Potential sources of error may include:
 - Navigation errors – failure to locate functions, excessive keystrokes to complete a function, failure to follow recommended screen flow.
 - Presentation errors – failure to locate and properly act upon desired information in screens, selection errors due to labeling ambiguities.
 - Control usage problems – improper toolbar or entry field usage.
- Exercise the application or web site under controlled test conditions with representative users. Data will be used to assess whether usability goals regarding an effective, efficient, and well-received user interface have been achieved.
- Establish baseline user performance and user-satisfaction levels of the user interface for future usability evaluations.

EVALUATION

- Challenges faced- identify snags or problems encountered.
- Constraints are factors that limit the project's options, restrict performance, affect scheduling, or limit the project in some fashion.
- Lessons learned how your topic related to your academic interests or personal interests.
- Give your perspective on the problem.
- Put your results in the context of other research.

CONCLUSION

- Summarize your results and restate their educational value

RUBRIC

Implementation Plan, Testing & Project Management

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Use of suitable project management to used throughout the project	- Project planning skills are very limited in application - Limited goal setting and use of strategies results in missed deadlines - In-sufficient information is accessed to implement strategies - Articulates the approach taken with limited evaluation	- Demonstrates a knowledge of, but is limited in application of project-planning skills. - Goals and strategies are identified but are not employed to consistently meet deadlines - Identifies information and resources to implement strategies - Partially evaluates the effectiveness of the approach.	- Demonstrates an adequate understanding of and applies standard project-planning skills. - Establishes goals and the strategies to meet those goals in a timely manner. - Uses information and resources to implement strategies - Adequately evaluates the effectiveness of approach.	- Demonstrates a broad understanding of and applies innovative project planning skills - Establishes goals and personalizes strategies to meet those goals, often in advance of deadlines - Acquires and strategically uses information and resources to implement strategies - Evaluation of approach provides new insights to project management and learning - Includes tracking documents such as schedule, issue log, change log, risk management plan, communication plan
Implementation Plan	No development plan or completely irrelevant plan	The Implementation plan does not address every step of development and does not create clear actionable tasks.	The Implementation plan has several reasonable steps for development, but they are not clearly defined or do not cover all aspects of development	The development plan has many distinct, logical steps that give a clear path for development. Explanation of tools to be used to support implementation

Prototype scope	No documented analysis on the scope of the prototype's functional or non-functional requirements/features, interfaces, interaction, reports and algorithms	Limited documented analysis on the functional requirements/features that will be developed as part of the prototype Limited documented analysis on the non-functional requirements/features that will be developed as part of the prototype Limited documented analysis on the scope of the prototypes interaction /interfaces / navigation with different stakeholders Limited documented analysis on the scope of the prototype's reports Limited documented analysis on the scope of the prototype's automated triggers /algorithms	Some documented analysis on the functional requirements/features that will be developed as part of the prototype Some documented analysis on the non-functional requirements/features that will be developed as part of the prototype Some documented analysis on the scope of the prototypes interaction /interfaces / navigation with different stakeholders Some documented analysis on the scope of the prototype's reports Some documented analysis on the scope of the prototype's automated triggers /algorithms	Well documented analysis on the functional requirements/features that will be developed as part of the prototype Well documented analysis on the non-functional requirements/features that will be developed as part of the prototype Well documented analysis on the scope of the prototype's interaction /interfaces / navigation with different stakeholders Well documented analysis on the scope of the prototype's reports Well documented analysis on the scope of the prototype's automated triggers /algorithms
User Evaluation Plan: Completeness	No plan or irrelevant plan.	The Prototype evaluation plan is unclear or lacks significant details	The Prototype evaluation plan is designed to produce some useful data, but is not justified by the student (e.g. why are you doing what you are doing?- why 6 stakeholders? Why in a school? etc).	The Prototype evaluation plan is very clearly motivated or innovative in a way that will ensure rich and interesting data to address the design goals.
User Evaluation Plan: Appropriateness	No plan or irrelevant plan.	The evaluation plan exists, but is minimal, unclear, and is not well thought out.	The evaluation plan is mostly complete, but does not cover all questions about stakeholder feedback thoroughly (what will be tested, what you want to learn from the stakeholder, when, where, participants).	The evaluation plan is complete, to answer all questions specifically, and shows a clear process for stakeholder feedback.
Testing: Plan	Testing methodologies include no black box and white box testing areas.	Testing methodologies include limited black box and white box testing areas.	Testing methodologies include most black box and white box testing areas.	Testing methodologies include all black box and white box testing areas.

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Wireframe / Storyboarding Prototype #1 (low-fidelity) Header, footer, navigation, content objects, branding elements, and their respective locations Groupings of elements such as side bars, navigation bars, and content areas Labelling, page title, navigation links Placeholders, content text and images.	No prototype or completely irrelevant prototype.	The prototype is not interactive or interactions are broken, and there is no defined purpose for many elements in the prototype.	The prototype is mostly complete, although there are some functions that are not yet interactive, some elements have no defined purpose, and it can be difficult to know how to use certain parts of the prototype.	The prototype has enough detail that the user can see all of the interactions, understand how they work, and a programmer could use the prototype to create a functional application with a defined flow.
Usability Report	Report is very poorly presented, has many spelling or grammar errors, does a very poor job of assessing the usability strengths and shortcomings of the prototype, and makes very few good recommendations for improving the prototype.	Report is poorly presented, has some spelling or grammar errors, does a poor job of assessing the usability strengths and shortcomings of the prototype, and makes few good recommendations for improving the prototype.	Report is professionally presented, has no spelling or grammar errors, does a good job of assessing the usability strengths and shortcomings of the prototype, and makes relatively good recommendations for improving the prototype.	Report is professionally presented, has no spelling or grammar errors, does an exceptionally good job of assessing the usability strengths and shortcomings of the prototype, and makes intelligent recommendations for improving the prototype.

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Quality of Stakeholder Critique	No critique provided from stakeholders of the project.	Critique was conducted on time, but includes equal or less than 2–3 positives and 1–2 criticisms.	Critique was conducted on time, includes at least 3–4 positives and 2–3 criticisms, critique is focused exclusively on the design solution, critique provides real recommendations for improvement.	Critique was conducted on time, includes at least 4–5 positives and 3–4 criticisms, critique is focused exclusively on the design solution, critique provides real recommendations for improvement.
Navigational Skeleton	No home screen or navigational skeleton, no URL	Home screen has little content, and navigation does not work. OR, the student submitted a prototype URL, but the prototype wasn't viewable.	Home screen appears to have most of its content, and the major navigations are present.	Home screen and navigational skeleton are very thorough and well planned. The navigational skeleton gives a real feel for the flow of the application and is clearly thought through.
Clarity of Diagrams and level of abstraction	Labels are often. No abstraction	Labels are somewhat cryptic. Limited level of abstraction	Diagrams are mostly well labeled. Diagrams are generally at an appropriate level of abstraction, though a stakeholder familiar with the problem domain might need some guidance to understand them	Diagrams are well labeled and at an appropriate level of abstraction (e.g. 0-level DFD, 1-level DFD, and 2-level DFD) so that stakeholders familiar with the problem domain could readily understand them.
Correct Use of Notation	Diagrams use notation inappropriate to the diagram type or contain a large variety of errors in the application of the notation	Some notation used in the diagrams are appropriate to the diagram type. Several sorts of errors are made in the application of each diagram type.	Most notation used in the diagrams are appropriate to the diagram type. At most two sorts of errors are made in the application of each diagram type.	All notation used in the diagrams are appropriate to the diagram type and is used correctly.
Display-Action Response Model	Flaws in the prototype seriously undermine its effectiveness Prototype missing	Prototype capture the key design features but minor details are missing or are provided for limited amount of screens	All the features in the design are described to some degree for most screens	The prototype unambiguously describes how each aspect of the design works for all screens

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Pseudocode / Algorithm: Readability	Pseudocode/ Algorithm is difficult to understand and is unorganized.	Pseudocode/ Algorithm is somewhat understandable, but very little of the code is organized logically.	Pseudocode/ Algorithm is clear, understandable, and the code is organized into steps, though there may be a few extra steps.	Pseudocode/ Algorithm is very clear, understandable, readable, and organized into as few steps necessary as possible. Includes all triggers, preconditions, post conditions, normal flow, alternative flows, exception flows and integrations.
Pseudocode / Algorithm: Demonstrated ability to convert to a program.	Code / algorithm was not converted into a program.	The code / algorithm was limited when converted into a program	The code/ algorithm was somewhat converted into a program	The code/ algorithm was easily converted into a program.

Prototyping, Project Report Updates (Enterprise Environmental Factors and Issues, Conclusion)

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Prototype #2 (high-fidelity)	No second prototype or completely irrelevant prototype.	The prototype is not interactive or interactions are broken, and there is no defined purpose for many elements in the prototype.	The prototype is mostly complete, although there are some functions that are not yet interactive, some elements have no defined purpose, and it can be difficult to know how to use certain parts of the prototype, and/or it does not diverges from the first prototype and explore different interfaces implementing the same idea.	The prototype has enough detail that the user can see all of the interactions, understand how they work, and a programmer could use the prototype to create a functional application with a defined flow. The prototype clearly diverges from the first prototype and explores different interfaces implementing the same idea
Interactive Prototype	No prototype or irrelevant prototype.	The prototype is not interactive, lacks many features, and has many bugs; the design does not work with the goal. OR, the student submitted a prototype URL or app, but the prototype wasn't viewable	The prototype is mostly interactive, with only a few features missing and only one or two bugs; the design accomplishes the minimum requirements of the goal.	The prototype is completely interactive accessible via URL/or app, reflects the feel of the final prototype, and is ready for user testing; the design accomplishes the entire goal. Multiple features present in the interface; Well thought out feature set

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Revised Prototype	The revised prototype does not effectively incorporate the feedback and data gathered in the usability test, or is not a significant improvement over the prior iterations because the test was poorly executed resulting in flawed data.	The revised prototype incorporates some of the feedback and data gathered in the usability test, but the prototype does not exhibit significant improvements in the design solution.	The revised prototype incorporates the feedback and data collected in the usability test resulting in a somewhat improved interface design solution.	The revised prototype incorporates the feedback and data collected in the usability test resulting in a more appropriate and usable interface design solution.
List of Changes	No changes or completely irrelevant changes.	The student only identified a few changes from the evaluation feedback and a large amount of feedback is ignored in the new prototype	The student made many of the simpler suggestion changes, but some of the more complex or difficult issues were not addressed	The student made several insightful and specific changes based on the evaluation feedback.

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Login and registration Interface / dashboard complexity	Prototype contains online registration and login for 1 type of stakeholder.	Prototype contains online registration and login for 2 types of stakeholders each has access to interact with different dashboards/interfaces/data. Overall screen design –limited Consistent style used Screen layout –limited use of menus, buttons, positioning of fields, and so on Screen navigation – not easy navigation between screens Software functions – limited functions works Data validation – limited appropriate validation of user-entered data.Type and range etc.	Prototype contains online registration and login for 3 types of stakeholders each has access to interact with different dashboards/interfaces/data. Overall screen design –good Consistent style used Screen layout –good use of menus, buttons, positioning of fields, and so on Screen navigation – somewhat easy navigation between screens Software functions – most works well Data validation – some appropriate validation of user-entered data.Type and range etc.	Prototype contains online registration and login for 4+ types of stakeholders each has access to interact with different dashboards/interfaces/d ata. Overall screen design – well thought out Consistent style used Screen layout –excellent use of menus, buttons, positioning of fields, and so on Screen navigation –easy navigation between screens Software functions –all works well Data validation – appropriate validation of user-entered data.Type and range etc. Forms –opportunity for auto-filling of some fields

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Report tables	Prototype includes no Report tables	Prototype includes 1-2 Report tables contain both information about the report as a whole, such as the report name or decisions made from the report, and field-level information, such as which is data fields are displayed and any calculations.	Prototype includes 3-4 Report tables contain both information about the report as a whole, such as the report name or decisions made from the report, and field-level information, such as which is data fields are displayed and any calculations.	Prototype includes 5+ Report tables contain both information about the report as a whole, such as the report name or decisions made from the report, and field-level information, such as which is data fields are displayed and any calculations. Common attributers of the table include: name, description, decisions from the report, objectives, audience, trigger, data fields, data volume, frequency, display format and calculation.
Coding Concepts	Some connection between app functionality and underlying code.	Describes the specific coding tasks necessary to build their app; demonstrates how that code powers the app's functionality.	Describes the specific coding tasks necessary to build their app; demonstrates how that code powers the app's functionality.	Explains the app's architecture, data structure, algorithms and features; discusses decision making in developing this approach.. Code is well documented with comments

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Effective Use of HTML Prototyping Methods	Prototype was not completed on time, does not show multiple states or pages, contains no interactivity.	Prototype was completed on time, shows multiple states and/or pages, but interactivity is limited to "slap and map" model.	Prototype uses HTML and CSS. Prototype was completed on time, shows multiple states and/or pages, moderate interactivity allows users to move through the experience, Ajax style transitions are faked with multiple pages.	Prototype uses HTML, CSS, and JavaScript. Prototype was completed on time, shows multiple states and/or pages, moderate interactivity allows users to move through the experience.
Correspondence of code and design	Code for the system is inconsistent with design diagrams apart from a few minor discrepancies	Code for the system is somewhat consistent with design diagrams apart from a few minor discrepancies.	Code for the system is mostly consistent with design diagrams apart from a few minor discrepancies.	Code for the system is consistent with design diagrams, both structurally (as documented by Design Class Diagrams) and behaviorally (as documented by Interaction Diagrams).
Testing: Outputs	Test cases are absent Unit testing is not applied	Test cases identified are not clear Unit testing is somewhat applied for all areas of the prototype	Test cases identified are clear, but may not be realistic. Unit testing is mostly applied for all areas of the prototype	Test-cases identified are clear and realistic. Unit testing is applied for all areas of the prototype

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Logical flow of ideas and outputs between report section (structure of the report)	Flow of ideas and outputs between report sections are absent.	Flow of ideas and outputs between report sections are lacking.	Flow of ideas and outputs between report sections are somewhat present.	Chapters well-organized/constructed and report sections fit together.
Spelling, grammar and Syntax Clarity and style	Frequent errors- Punctuation, misuse of words, spelling, sentence construction make the work largely incomprehensible Style makes work largely incomprehensible. A self-centered ramble evoking emotive terms.	Many errors – punctuation, misuse of words, spelling, and sentence construction occasionally make meaning unclear. Style too colloquial or makes work difficult to understand. Lacks objectivity.	Acceptable standard. Few errors, typos. Effective use of English. Clear and easy to read. Content free from generalizations and unsupported statements and opinions.	Very high standard. Negligible errors. Very effective use of English. Clear, concise and easy to read. Impartial academic style.
Political Factors	Did not include any Political factors and did not understand how they affect the project, prototype and/ or organization/stakeholders	Included Political factors but only emphasized how 2-3 of the elements affects the project, prototype and/ or organization/stakeholders	Gave at least 4 Political factors but did not explain each one or the impact they have on the project, prototype and/ or organization/stakeholders	Included at least 4 Political factors and emphasized how each element affects either the project, prototype and/ or organization/stakeholders .

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Economic Factors	Did not include any Economic factors and did not understand how they affect the project, prototype and/ or organization/stakeholders	Included Economic factors but only emphasized how 2-3 of the elements affects the project, prototype and/ or organization/stakeholders	Gave at least 4 Economic factors but did not explain each one or the impact they have on the project, prototype and/ or organization/stakeholders	Included at least 4 Economic factors and emphasized how each element affects either the project, prototype and/ or organization/stakeholders .
Social Factors	Did not include any Social factors and did not understand how they affect the project, prototype and/ or organization/stakeholders	Included Social factors but only emphasized how 2-3 of the elements affects the project, prototype and/ or organization/stakeholders	Gave at least 4 Social factors but did not explain each one or the impact they have on the project, prototype and/ or organization/stakeholders	Included at least 4 Social factors and emphasized how each element affects either the project, prototype and/ or organization/stakeholders .
Legal / Ethical Factors	Did not include any Legal / Ethical factors and did not understand how they affect the project, prototype and/ or organization/stakeholders	Included Legal / Ethical factors but only emphasized how 2-3 of the elements affects the project, prototype and/ or organization/stakeholders	Gave at least 4 Legal / Ethical factors but did not explain each one or the impact they have on the project, prototype and/ or organization/stakeholders	Included at least 4 Legal / Ethical factors and emphasized how each element affects either the project, prototype and/ or organization/stakeholders .

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Stated Lesson Learned	Didn't state what he/she learned from facing this challenge.	Stated the lesson learned but didn't explain what he/she learned from experiencing this challenged.	Stated the lesson learned but somewhat explained what he/she learned from experiencing this challenged.	Stated the lesson learned and explained what he/she learned from experiencing this challenged. Stated how they feel now after overcoming this challenge and stated strategies they will use to face future challenges. Reflect on whether or not you met the project success criteria (good and bad). Described what will be done differently on the next project based on experience working on this project

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Evaluation (of both prototype and project)	Lacks objectivity. Only minor relevant evaluation of the work is presented. Limited evaluation without clear links to the objectives.	Some evaluation with some links to work undertaken. Many key issues identified.	Significant evaluation of the outcome (or product) with little emphasis on the process and methods. Clearly stated evaluation firmly based on evidence provided. Feasible set of recommendations linked with project objectives.	Reflective and insightful evaluation. Assessment of both project and prototype. Choices of approaches and methods re-visited in light of outcomes. Objectives fully reviewed. Clear understanding of potential and limitations. Appropriate and realistic recommendations consistent with results. Clear indication of how the project might be taken forward further after the prototype for full implementation.

Dimension	Unsatisfactory	Emerging	Proficient	Exemplary
Conclusion	No recognizable conclusion.	Conclusion does not do justice to the report. Inadequate summary of key points/ issues/ outcomes. Conclusions drawn but insubstantial or unproven.	Good. Summarizes key points/ issues and outcomes. Conclusions drawn, supported by body of writing.	Excellent. Accurately identifies and summarizes key points/ issues and outcomes with reference to the original problem. Critical, reflective conclusions. Propose new work in light of findings. A convincing argument shows that the recommendation is superior to the usual alternatives. Discussion of limits and reservations (and alternatives obvious in retrospect).

RUBRIC FOR ASSESSING PROTOTYPE DEMONSTRATION.

CRITERIA	PERFORMANCE LEVELS			Notes	Points
	10 pts - Excellent	7 pts – Acceptable	2 pts – Needs Improvement		
Overall Appearance	The prototype is very well put together. The effort put into making the model is clearly apparent. Aesthetics were thoroughly considered and prototype displays craftsmanship and proficient use of available tools and resources.	The prototype is assembled properly. The effort put into the prototype is satisfactory. Aesthetics were considered, however there are areas where lines, edges, and components could be tidied up or better organized. There is a satisfactory use of tools and resources to complete the prototype.	The prototype is poorly built and falls apart. The prototype lacks apparent effort and consideration of aesthetics. Lines are rough, edges do not meet, and components are not arranged in an organized manner. Prototype demonstrates a lack of proficiency with tools and resources.		
Presentation Delivery	(1) Speak clearly and fluently. (2) Communicate and explain ideas clearly. (3) Organized and structured presentation. (4) Confident and proud of the work. (5) Slides cover complete, accurate description of important outcomes.	(1) Speak legibly. (2) Communicate and explain ideas adequately. (3) Somewhat organized and structured presentation. (4) Not very confident and proud of the work. (5) Slides cover accurate description of most of important outcomes.	(1) Hard to understand: weak language, low voice, mumbled speech, etc. (2) Unobvious explanation of ideas. Requires discussion to clarify. (3) Unclear structure of presentation. (4) Doubtful of the work. (5) Slides cover some of the outcomes.		

Functional Demo	(1) prototype present and working perfectly. Prototype is fully functional and meets the criteria and constraints of the project. Prototype demonstrates maximal effort with little to no compromise of the aesthetics of the product.	(1) prototype present and partly working. Prototype is partially functional and meets many, but not all, the criteria and constraints of the project. Prototype demonstrates satisfactory effort, however the functionality comes at the expense of aesthetics.	(1) Solution present but not functional. Prototype is not functional and does not meet the criteria or constraints of the project. Prototype demonstrates little effort and neither functionality nor aesthetics are achieved.		
Working Output	Student's design was difficult and finished the prototype with a great amount of effort and problem solving. Prototype demonstrates a commitment inside and outside of the classroom. Demonstrates a significant improvement upon mockup design.	Student put forth a good amount of effort but the prototype still looks like it could have a little more time spent on it. Prototype demonstrates that prototype was worked on during class time, only. Demonstrates a moderate improvement upon mockup design.	There was no effort apparent in putting together the prototype. It was constructed in the least amount of time with easy construction practices. Prototype demonstrates that time inside and outside of the classroom was not spent efficiently. Demonstrates little to no improvement upon mockup design.		
Creativity/ Innovation	Completely new idea or significant improvement upon existing technology. Innovative. Prototype demonstrates a sophisticated use of the engineer's design process to develop a truly unique solution.	Minor improvement on existing technology. Moderately Innovative. Prototype demonstrates the satisfactory use of engineer's design process to develop or embellish upon technology.	No improvement upon existing technology. Lacking innovation. Prototype demonstrates that the engineer's design process was not used.		



Total					

SAMPLE MARK SHEET

STUDENT NAME: _____

STUDENT NUMBER: _____

CRITERIA	Lecturer's Mark	Moderator's Mark	Total Mark
Submit Solution Implementation Plan			10
Meet with Faculty Advisor on 1st Solution Prototype			10
Submit 2nd Solution Prototype and 3rd Draft Report			20
SUBMIT FINAL PROJECT			20
FINAL SOLUTION DEMONSTRATION			40



GRAND TOTAL			100
	GRADE		
			DATE

REFERENCING

The MLA (Modern Language Association) style is most commonly used to write papers and cite sources within the liberal arts and humanities and is the approved method for presentation of written work at UWI-ROYTEC. Examples for the general format of MLA research papers, in-text citations, endnotes/footnotes, and the Works Cited page can be obtained at <http://owl.english.purdue.edu/owl/resource/747/1> or MLA Handbook for Writers of Research Papers (7th ed.) and the MLA Style Manual and Guide to Scholarly Publishing (3rd ed.), which are available in the Library. Students may also refer to the text used in the course ENGL 0102-Written Communication, Kolin, Philip. *Successful Writing at Work*. Boston: Houghton-Mifflin., 2004.

Students are reminded that direct quotations, facts and figures, as well as ideas and theories, from both published and unpublished works must be referenced. Referencing is a standard method of acknowledging sources of information and ideas that have been used in assignments in a way that uniquely identifies their source. Penalties for failure to reference your work will be applied under UWI-ROYTEC's Policy on Plagiarism.

General Guidelines

- Type your paper on a computer and print it out on standard, white 8.5 x 11-inch paper.
- Double-space the text of your paper, and use a legible font (e.g. Times New Roman). Whatever font you choose, MLA recommends that the regular and italics type styles contrast enough that they are recognizable one from another. The font size should be 12 pt.
- Leave only one space after periods or other punctuation marks (unless otherwise instructed by your instructor).
- Set the margins of your document to 1 inch on all sides.
- Indent the first line of paragraphs one half-inch from the left margin. MLA recommends that you use the Tab key as opposed to pushing the Space Bar five times.
- Create a header that numbers all pages consecutively in the upper right-hand corner, one-half inch from the top and flush with the right margin. (Note: Your instructor may ask that you omit the number on your first page. Always follow your instructor's guidelines.)
- Use italics throughout your essay for the titles of longer works and, only when absolutely necessary, providing emphasis.
- If you have any endnotes, include them on a separate page before your Works Cited page. Entitle the section Notes (centered, unformatted).

Formatting the First Page of Your Paper

- Do not make a title page for your paper unless specifically requested.
- In the upper left-hand corner of the first page, list your name, your instructor's name, the course, and the date. Again, be sure to use double-spaced text.
- Double space again and center the title. Do not underline, italicize, or place your title in quotation marks; write the title in Title Case (standard capitalization), not in all capital letters.
- Use quotation marks and/or italics when referring to other works in your title, just as you would in your text: *Fear and Loathing in Las Vegas* as Morality Play; Human Weariness in "After Apple Picking"
- Double space between the title and the first line of the text.
- Create a header in the upper right-hand corner that includes your last name, followed by a space with a page number; number all pages consecutively with Arabic numerals (1, 2, 3, 4, etc.), one-half inch from the top and flush with the right margin. (Note: Your instructor or other readers may ask that you omit last name/page number header on your first page. Always follow instructor guidelines.)

Excerpt taken from: <https://owl.english.purdue.edu/owl/resource/747/01/> [Accessed on May 7, 2015]

Excerpts taken from: The PMI Guide to Business Analysis (2017)

Excerpts taken from: Business Analysis for Practitioners: A Practice Guide (2015)

Excerpts take from: Systems Analysis and Design, 9th Edition, Gary B. Shelly; Harry J. Rosenblatt, ISBN-10: 0538481617 | ISBN-13: 9780538481618

Wishing you all the best in your
Senior Seminar II!